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Calibration of an optical particle counter to provide $PM_{2.5}$ mass for well-defined particle materials

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Abstract

A light scattering optical particle counter (OPC) was re-calibrated for a well-defined particle material, to convert optical equivalent size both into aerodynamic and volume equivalent diameters. For specific test dusts, the calibrated OPC provides a direct read-out dust mass vs. aerodynamic diameter and — by weighting the appropriate size bins with the standard transmission curve — as total $PM_{2.5}$ mass. The time resolution is in the order of 1 s.

The paper gives details of an accurate calibration method for supermicron particle materials where monodisperse size standards are generally unavailable. The procedure is based on measuring the transmission curve of an aerodynamic collection device in parallel with the optical particle counter and an aerodynamic particle sizer.

The calibration was verified in more than 100 tests, where particle emissions from bag-house filter media were recorded in parallel with the recalibrated OPC and a $PM_{2.5}$ cyclone (BGI SCC). The deviation between OPC and SCC is less than 20% for about 95% of the results, with no systematic deviation.

Keywords: $PM_{2.5}$; Particle emissions; Cleanable filter media; Optical particle counter

1. Introduction

The adoption of PM_{10} and $PM_{2.5}$ criteria for ambient air in the US and EU is also driving a reevaluation of air cleaning devices with regard to their emission characteristics. In many cases this requires a considerable departure from past thinking, which was based entirely on mass concentrations of emitted dust. This is also the case for cleanable filter media (so-called "bag" or "pocket" filters), which are used worldwide for emission control of high volume sources carrying high dust loads on the order of g/m^3, such as power plants, waste incineration plants or cement mills. Bag filters operate cyclically and therefore require periodic cleaning by pressure pulse or similar techniques. The filters are commonly made of dense needle felts designed to work as surface barrier filters, and hence at relatively low total emissions when compared to other control devices such as ESPs. In recent years, much work has also been done with ceramic and metallic media for high temperature applications, which are even more efficient (Umhauer, Berbner, & Hemmer, 2000).

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Current standards for the characterization of needle felts (ASTM D 6830-02, 2002; JIS Z 8909-1, 2005; VDI/DIN 3926, 2004) are based on a chemically uniform test dust. Emissions are determined on the basis of total emitted mass concentration by gravimetric evaluation of filter samples. In addition, options are provided or being considered to measure $PM_{2.5}$ on the basis of a classical aerodynamic/gravimetric method such as cyclones (Binnig, Meyer, & Kasper, 2005; Hering, Kreisberg, & John, 2003; Kenny, Gussman, & Meyer, 2000) or impactors (such as in ASTM D 6830-02). Unfortunately, the emissions from such filter elements vary strongly with time on a scale of typically a few minutes. In addition, the emissions of modern needle felts are very low. Thus, it is becoming increasingly difficult to accumulate enough mass over a few filter cycles for a meaningful measurement by traditional gravimetric methods, let alone to resolve the emission behavior from cycle to cycle.

Alternatives with higher sensitivity in the size range above 1 μm include the aerodynamic particle size (APS) and the TEOM (ASTM D 6830-02). The APS would also require an additional, dust-specific calibration to convert particle number to particle mass, and it has a rather limited operation range in terms of concentration. The TEOM can detect total mass with good sensitivity but in itself has no capability to resolve particle size. Consequently, there are many arguments in favor of light scattering optical particle counters (OPC) to do mass-based emission characterization, including very good size and time resolution in the particle size range of interest. Aside from sensitivity, time resolution is especially interesting for testing pulse cleaned filter media, because particles are emitted in bursts immediately following regeneration, and because the aging process can be resolved.

Despite this potential, there is a significant drawback with OPCs, due to the fact that optical sizes and number concentrations are not easily converted into aerodynamic diameters and particle mass required for most standards. Fortunately, this is less of a handicap in testing cleanable filters than for environmental sampling, because needle felts are typically tested with standardized inorganic test dusts of uniform composition. For a chemically and physically well-defined particle material it is possible, in principle, to calibrate an OPC to read mass directly (Friehmelt & Heidenreich, 1999). Recently, Binnig et al. (2005) successfully integrated an OPC into a test setup for pulse cleaned filter media, to monitor particle emissions in terms of particle mass vs. aerodynamic diameter, and in terms of total $PM_{2.5}$ dust mass emitted. It should be noted, however, that some OPCs are less suitable for converting number concentrations into mass, because they cannot resolve the size range around 1 μm (near the most-penetrating particle size MPPS), due to the shape of their response curve.

The objective of the current paper is to describe the detailed approach taken to convert OPC size spectra in the size range of typically 0.3 to 10 μm into $PM_{2.5}$ mass for a known particle material. We will provide details of calibration and experimental procedures to resolve certain practical difficulties (such as the lack of monodisperse particle fractions for the test dust in question) and present comparisons with the conventional aerodynamic/gravimetric technique, in order to give an impression of the accuracy achieved.

2. Method of converting OPC data into $PM_{2.5}$ dust concentrations

Using a light scattering OPC (of sufficient resolution) to measure $PM_{2.5}$ concentrations requires three conversion steps: the optical equivalent particle size d_{opt} has to be converted into an aerodynamic equivalent size d_{AE}; the particle number in a given size class has to be converted into a particle mass by converting d_{opt} into a volume equivalent size, d_{VE}; and finally the concentration in a given size class has to be weighted with the nominal $PM_{2.5}$ transmission curve (USEPA, 1997), which is also a function of the aerodynamic diameter d_{AE}. The total PM mass is then obtained by summing up the mass in all size classes.

Since neither d_{VE} nor d_{AE} are measured directly by the OPC, it is necessary to obtain the necessary conversion curves from d_{opt} by calibration with a particle material which is well defined in terms of density and refractive index. A separate calibration is required for each material. This issue is complicated by the fact, that sufficiently monodisperse size fractions are impossible to obtain for the predominately supermicron particle materials used in filter testing in the size range of about 0.5 to 10 μm.

The approach taken here was to re-measure the transmission curve of an aerodynamic separator (in this case a cyclone) with accurately known $PM_{2.5}$ characteristic with the OPC. Since the cyclone has a unique relationship between particle size and efficiency, one can assign an aerodynamic diameter d_{AE} to each value of optical diameter d_{opt} on the basis of the "optical" equivalent of the transmission curve. The correlation varies of course with particle density, shape factor and optical properties, and requires a separate calibration for each particle material. Once the conversion curve between d_{opt} and d_{AE} is available, it is easy also to convert to the Stokes diameter which is linked to d_{AE} by the well-known expression:

$$d_{AE} \cdot C(d_{AE})\rho_0^{1/2} = d_{ST} \cdot C(d_{ST})\rho^{1/2}, \tag{1}$$

where C denotes the slip correction factor and ρ the particle density. In the following, we shall use the calibration factor $d_{ST,i}/d_{opt,i}$ for each size channel i of the OPC.

The conversion from number concentration c_n to mass concentration c_m in each of the discrete size channels of the OPC (in our case 64) can be done by using the simple formula:

$$c_{m,i} = c_{n,i} \cdot \frac{\pi}{6} \cdot d_{VE,i}^3 \cdot \rho, \tag{2}$$

where i is the channel number. Although there is generally no straightforward theoretical relationship between d_{VE} and d_{opt}, we know that d_{VE} is related to the Stokes diameter d_{ST} by

$$d_{VE} \cdot C(d_{VE}) = f \cdot d_{ST} \cdot C(d_{ST}), \tag{3}$$

where f is the dynamic shape factor (Kasper, 1982), which remains to be determined. Since f is expected to be ≈ 1 for many mineral powders, we can assume that $C(d_{ST}) \approx C(d_{VE})$ and therefore drop the slip correction in (3). Since the link between d_{opt} and d_{ST} has already been established from the "optical" transmission curve as described above, we see that the conversion between d_{opt} and d_{VE} can in fact be based on the same calibration.

Combining Eqs. (1)–(3), one thus obtains a useful expression for the desired conversion between particle mass concentration and particle optical diameter:

$$c_{m,i} = c_{n,i} \cdot \frac{\pi}{6} \cdot \left(d_{opt,i} \cdot \frac{d_{ST,i}}{d_{opt,i}}\right)^3 \cdot f^3 \cdot \rho. \tag{4}$$

The factors f^3 and ρ can be lumped together into a mass conversion factor *MCF*:

$$MCF = f^3 \cdot \rho, \tag{5}$$

which should be reasonably independent of particle size, provided we are dealing with an uniform calibration material (which is generally the case for the powders used to test cleanable filter media, as opposed to cocktails such as ISO test dust). *MCF* (and hence f) can then be obtained easily by comparing the sum $\sum_{c_{m,i}}$, with the total particle mass collected on a back-up filter downstream of the OPC.

The last step on the way to measuring total PM mass with an OPC requires multiplying each channel with the appropriate weighting factor $P_{PM,i}$ to emulate the nominal aerodynamic $PM_{2.5}$ transmission curve. Since this factor is a function of d_{AE}, it can be assigned to each OPC channel on the basis of the calibration $d_{ST,i}/d_{opt,i}$. We thus finally obtain for each size class

$$c_{m,i} = \frac{c_{n,i}}{E_{c,i}} \cdot \frac{\pi}{6} \cdot \left(d_{opt,i} \cdot \frac{d_{ST,i}}{d_{opt,i}}\right)^3 \cdot MCF \cdot P_{PM,i}. \tag{6}$$

Here we have introduced yet another correction to account for the actual counting efficiency $E_{c,i}$ curve of an OPC.

3. Experimental setup and procedures

The *particle material* used for the OPC calibration was an alumina monohydrate powder known internationally by its trade name Pural SB (Manufacturer: Sasol). Pural SB is relatively free-flowing and commonly used to test cleanable filter media. It is produced in batches of several tons which typically last for a few years, during which one can work with a well-defined and stable particle size distribution. The particle mass size distribution contains a significant fraction of rather large particles (mass mean particle size MMPS $\approx 30\,\mu m$).

Nevertheless the MPPS of most needle felts is covered and the distribution reaches well into the size range relevant in case of damaged or leaking media.

The *optical particle counter* used for the emission measurements has a size range of 0.3 to 20 µm optical equivalent diameter (as determined by calibration with PS latex) with 64 size bins. The device uses white light and operates at

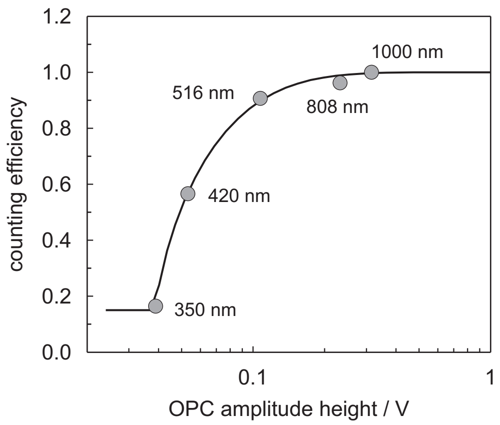

Fig. 1. Counting efficiency vs. OPC signal amplitude (photomultiplier acceleration voltage 872.5 V).

90° scattering angle, with an optically defined measuring volume. Hence it does not require introduction of the aerosol stream into a small optical cell, and the common problem of aerodynamic losses for large particles is minimized. On the other hand, only part of the sampling flow enters the measuring volume, but this has been carefully determined by calibration. It belongs to a family of devices developed in Karlsruhe which have been described in the literature before (Maus & Umhauer, 1996). The calibration curve of the OPC shows a slight change of slope around 1 μm , but does not flatten or invert. This is important for accurate measurements in the vicinity of the filter MPPS, which is typically around 0.5 to 0.8 μm for needle felts, corresponding to an aerodynamic diameter around 0.8 to 1.5 μm , depending on particle density. The device was typically operated with time resolutions of less than one second.

While the OPC has no aerosol inlet loss problem for particles above 1 μm , its counting efficiency near the lower end of the operating range can be expected to decrease, as for any OPC. If not accounted for, this deviation in the vicinity of the MPPS can cause serious errors for converting particle counts to mass. The counting efficiency was therefore determined with five latex fractions between 0.35 and 1 μm , by comparison with a calibrated condensation particle counter (CPC; TSI Model 3010). The latex particles were aerosolized with a Collison atomizer and then classified by a differential mobility analyzer (DMA) to remove empty residual particles which would be detected by the CPC. The number concentration was then determined simultaneously by the OPC and CPC. The CPC was checked periodically and known to be 100% efficient in the size range of interest. CPCs are generally more reliable than OPCs with respect to counting efficiency in the range just under 1 μm (Wen & Kasper, 1986).

The measured counting efficiency E_c is a function of the scattered light intensity and should therefore not be assigned directly to a fixed d_{opt} via the OPC calibration curve, without taking into account that the intensity of the light source decreases gradually during its operating life. Instead, the counting efficiency was correlated to the pulse amplitude A measured by the OPC, at a defined signal amplification, as shown in Fig. 1 for a fixed photomultiplier amplification voltage $U_{PM} = 872.5\,V$ and particle number concentrations $15\,cm^{-3} < c_n < 750\,cm^{-3}$ (coincidence error $\ll 1\%$). The counting efficiency E_c begins to drop significantly for pulse amplitudes below 0.1 V, which — depending on the light intensity — corresponds to latex equivalent diameters d_{opt} between 400 and 600 nm. For amplitude heights $A < 0.04\,V$ (corresponding to $300\,nm < d_{opt} < 380\,nm$), a constant value of 0.15 was assumed, since reliable measurements were not possible.

The curve shown in Fig. 1 is only valid for the U_{PM} used during the calibration experiment. Since the shift of the size calibration due to a decrease of light source intensity is compensated by an increase of U_{PM} (or an exchange of the light source, usually every 1000 h), pulse amplitudes measured with a U_{PM} different from 872.5 V are corrected with a factor shown in Fig. 2 before applying the efficiency correction function.

The *correlation of optical vs. aerodynamic particle diameter* was obtained with the help of a sharp-cut cyclone (SCC; manufactured by BGI, USA) with $PM_{2.5}$ cut-off characteristic at a flow rate of $1\,m^3/h$. The SCC was developed as a replacement for the WINS ("well impactor ninety-six") and is less sensitive to loading effects than the WINS (Kenny et al., 2000). This is also beneficial for our application, where comparison runs between SCC and OPC were conducted over long periods of time during filter tests.

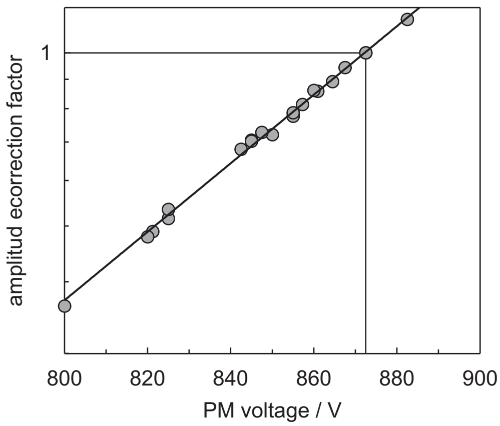

Fig. 2. Correction factor for the pulse amplitude height A when using a photomultiplier acceleration voltage different from 872.5 V.

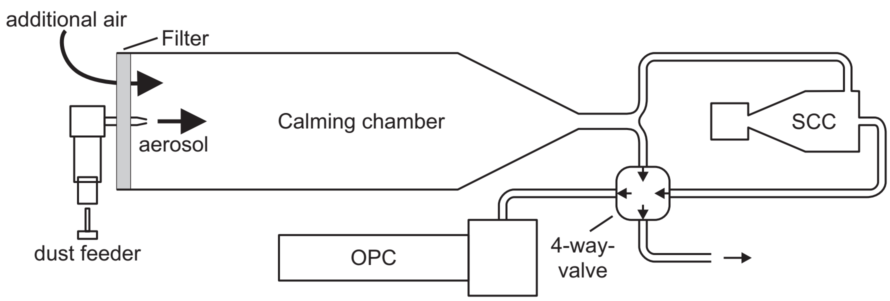

Fig. 3. Experimental setup for measuring the optical transmission curve of the SCC.

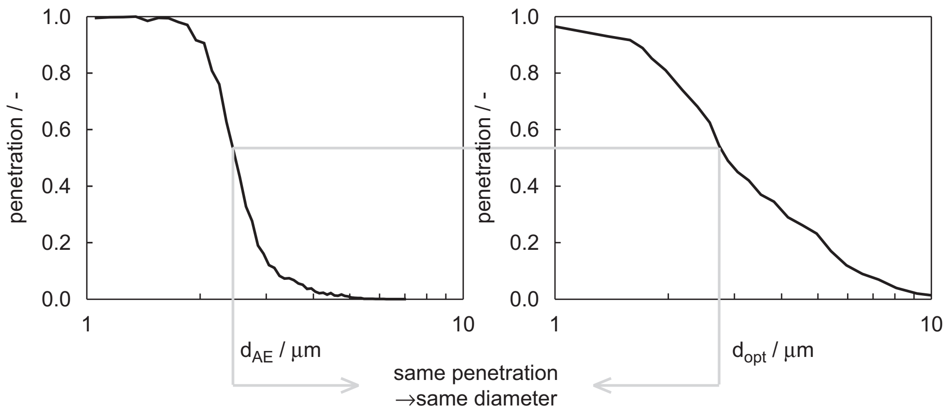

Fig. 4. Correlation between "aerodynamic" and "optical" transmission curves of the SCC cyclone, used to correlate d_{opt} with d_{AE}.

All calibration experiments were performed with the setup shown in Fig. 3, which is similar to the setup used by Maynard and Kenny (1995). The aerosol flow enters the calming chamber where it is mixed with the additional air. At the end of the calming chamber the air flow is split into two equal parts, of one which is fed directly into the SCC, the other into a 4-way valve. By switching the valve it is possible to alternate between upstream and downstream of the SCC in order to determine the optical transmission curve of the cyclone.

The conversion from d_{opt} to d_{ST} obtained by comparing the "optical" and "aerodynamic" transmission curve (Fig. 4) was limited to efficiencies between 0.2 and 0.8, to reduce statistical errors. For this reason, and since the

SCC transmission curve is quite steep ($P > 0.8$ for $d_{AE} < 2.1\,\mu m$ and $P < 0.2$ for $d_{AE} > 2.8\,\mu m$), a measurement at the standard flow cyclone rate ($1\,m^3/h$) covers only a small window of the entire size range of interest for testing surface filters. However, the aerodynamic transmission curve of the SCC can be shifted by changing the cyclone flow rate. The new curve can be recalculated with a formula by Buettner (1999) or measured. We have chosen the experimental approach using the setup described above, but replacing the OPC with an APS (TSI Model 3321) for flow rates from 0.8 to $3\,m^3/h$. It is known that the APS has a low counting efficiency in the size range of interest to us (Peters & Leith, 2003). However, the transmission curve we are interested in is determined as a series of concentration ratios (calculated separately for each size channel). Therefore any counting efficiency error cancels out.

The *mass conversion factor MCF*

$$MCF = \frac{\boldsymbol{m}_{\mathrm{total}}/\dot{\boldsymbol{V}}_{\mathrm{filter}} \cdot \boldsymbol{t}_{\mathrm{sample}}}{\sum_i c_{n,i}/E_{c,i} \cdot \pi/6 \cdot (d_{\mathrm{opt},i} \cdot d_{\mathrm{ST},i}/d_{\mathrm{opt},i})^3} \tag{7}$$

was determined by comparing the aerosol number concentration and size distribution measured by the OPC with the total mass simultaneously sampled on an analytic filter. During later filter tests, this experiment was repeated for different particle size distributions to ensure that the *MCF* is not size dependent.

4. Results and discussion

Calibration of d_{opt} vs. d_{AE} or d_{ST}: Fig. 5 shows a direct comparison of the "aerodynamic" $PM_{2.5}$ penetration curve (i.e. the curve obtained when measuring efficiency vs. d_{AE} by APS) and the "optical" $PM_{2.5}$ penetration curve of the SCC for the standard particle material Pural SB. If the x-axis is taken to be in units of d_{AE}, then following any horizontal line for a given efficiency will yield the optical size d_{opt} corresponding to d_{AE}. Since the scale of the horizontal axis is logarithmic, it is obvious from the relative shape of the two curves that there can be no constant conversion factor between d_{opt} and d_{AE} or d_{ST}.

The combination of measurements at various cyclone flow rates from 0.8 to $3\,m^3/h$ leads to the complete conversion curve for Pural SB (Fig. 6). Note that Fig. 6 shows d_{opt} vs. d_{ST} (instead of d_{AE}) because d_{ST} is more convenient for the conversion to d_{VE} because the slip correction can be dropped (see Section 2). The strong size dependence of the conversion factor d_{opt}/d_{ST} — especially for the range below about $1\,\mu m$ — is difficult to explain, but occurs systematically also for other particle materials such as limestone dust (Microcalcilin) and a different fraction of alumina monohydrate (Pural NF), and when using other OPCs.

Mass conversion factor MCF: The *MCF* defined in Eq. (7) was determined by comparing the total particle mass obtained by the OPC with the mass on a back-up filter. The average value from a large number of filter experiments is $2.65\,g/cm^3$. The filters used had different MMPS values and hence each *MCF* value is based on somewhat different particle size distribution. A detailed analysis of the data showed that the *MCF* can be regarded as constant over the particle size range of interest, except near the lower detection limit of the OPC. A slight increase of the *MCF* in that

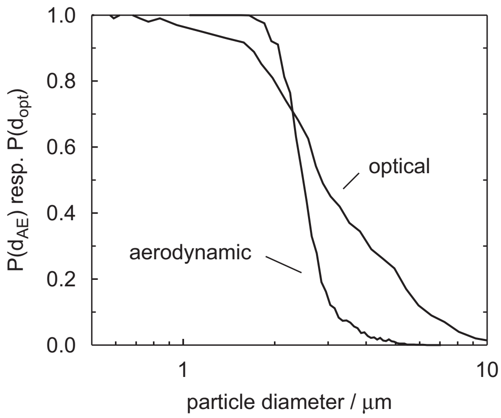

Fig. 5. Comparison of "aerodynamic" and "optical" cyclone penetration curves (Pural SB).

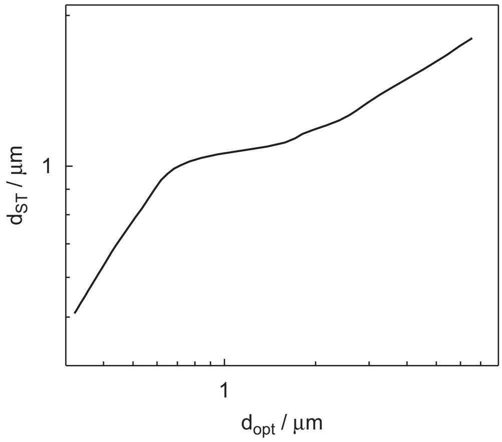

Fig. 6. Calibration curve between d_{ST} and d_{opt} (Pural SB).

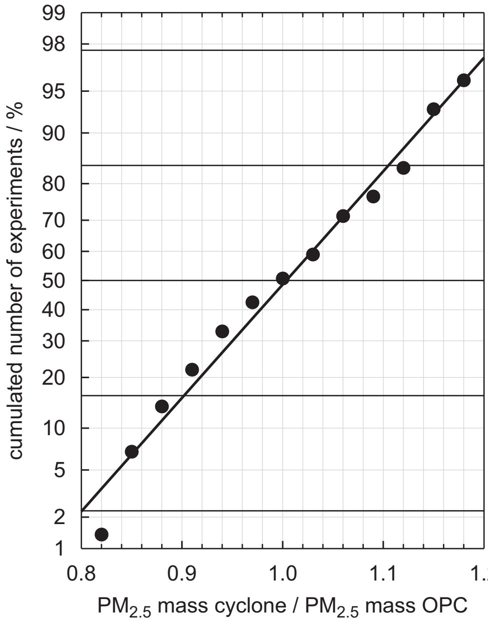

Fig. 7. Cumulative distribution of ratios of optically measured vs. cyclone based $PM_{2.5}$ mass concentration, obtained from various types of cleanable filter media.

region may be due to the fact that the OPC misses some of the emitted particles (due to the lower detection limit). However, this region does not contribute much to the total mass concentration.

By inserting the appropriate values for the *MCF* and for the particle density ($\rho = 2.78\,\mathrm{g/cm^3}$ for Pural SB) into Eq. (5), one can now also estimate the aerodynamic shape factor *f*. The derived value of 0.98 for *f* is typical for mineral dust particles.

Comparison with $PM_{2.5}$ data obtained by SCC: Nearly 100 filter experiments have been conducted so far with the same calibration functions programmed into the data evaluation software of the OPC, thereby giving a direct read-out of the $PM_{2.5}$ relevant emissions. In parallel to the OPC, the $PM_{2.5}$ concentrations were also captured in the conventional way, by using an SCC cyclone as preseparator followed by an absolute filter. Details of this experimental setup have been reported by Binnig et al. (2005).

This data set was used to calculate ratios of optical to aerodynamic mass, which are summarized in Fig. 7 in the form of a distribution. One sees that the results are very nicely approximated by a normal distribution around a ratio of 1 with a standard deviation of 0.10. In other words, 2/3 of the measurements are within $\pm 10\%$ of unity and 95% within $\pm 20\%$ of unity. Considering the differences in methodology and the number of conversion steps required for the OPC data, the agreement between the $PM_{2.5}$ mass measured by cyclone and that calculated on the basis of the OPC can be considered very good. It is especially interesting to note that there is no systematic deviation.

Outlook: Further improvements of the accuracy are possible but would require modifications of the way the OPC is designed to function. For example, aging of the light source with its attendant loss in intensity (and consequently also of signal intensity) is compensated at present by a gradual increase of signal amplification. This method preserves the size calibration of the OPC, but pushes a growing portion of the particle size distribution into channels with lower signal-to-noise ratios. Hence, there will be increasing statistical errors for small particles. A better method would be to gradually increase the supply voltage of the light source to maintain a constant intensity. This will minimize the influence of aging on the data quality, however, at the expense of a reduced life expectancy of the source.

Additional calibrations will be performed in the future for other particle materials including oil droplets. Especially the oil droplet data should give a better understanding of the calibration process, because the shape factor $f = 1$ and because it is more easily possible to produce monodisperse oil aerosols of known droplet size.

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